Employee Analyticss

Description

To analyzes employee, salary, and performance data to generate meaningful business insights.it evaluates promotion eligibility, attrition trends, hiring growth, and salary-performance patterns using advanced sql queries.overall, it demonstrates strong data analysis and data engineering skills through joins, aggregations, and analytical functions.

Structure

1.Master Table

```
create table Departments(
dept_id int primary key,
dept_name varchar(50) not null);

create table Locations(
location_id int primary key,
location_name varchar(50) not null);

create table Job_Roles(
role_id int primary key,
role_name varchar(50) not null,
min_salary decimal(10,2),
max_salary decimal (10,2));
```

2.Employee Table

```
create table Employees(
emp_id int primary key,
emp_name varchar(100) not null,
gender varchar(10),
dept_id int,
role_id int,
location_id int,
```

```
join_date Date,
status varchar(20) default 'Active',
foreign key(dept_id) references departments(dept_id),
foreign key(role_id) references job_roles(role_id),
foreign key(location_id) references Locations(location_id));
```

3.Salary Table

```
create table salaries(
salary_id int primary key,
emp_id int,
salary decimal(10,2),
effective_data date,
foreign key (emp_id) references employees(emp_id)
);
```

4.Performance Table

```
create table performance_reviews(
review_id int primary key,
emp_id int,
review_year int,
rating decimal(2,1) check (rating between 1 and 5),
foreign key(emp_id) references employees(emp_id)
```

Load Datasets

```
BULK INSERT Employees
FROM 'C:\SQLData\Employees_100_Records.csv'
WITH
(
   FIRSTROW = 2,
   FIELDTERMINATOR = ',',
   ROWTERMINATOR = '0x0a',
   TABLOCK,
   CODEPAGE = '65001'
);
```

```
BULK INSERT Departments
FROM 'C:\SQLData\Departments.csv'
WITH
(
   FIRSTROW = 2,
   FIELDTERMINATOR = ',',
   ROWTERMINATOR = '0x0a',
   TABLOCK
);

BULK INSERT Locations
FROM 'C:\SQLData\Locations.csv'
WITH
(
   FIRSTROW = 2,
   FIELDTERMINATOR = ',',
   ROWTERMINATOR = '0x0a',
   TABLOCK
);

BULK INSERT Job_Roles
FROM 'C:\SQLData\Job_Roles.csv'
WITH
(
   FIRSTROW = 2,
   FIELDTERMINATOR = ',',
   ROWTERMINATOR = '0x0a',
   TABLOCK
);

BULK INSERT performance_reviews
FROM 'C:\SQLData\performance_reviews.csv'
WITH
(
   FIRSTROW = 2,
   FIELDTERMINATOR = ',',
   ROWTERMINATOR = '0x0a',
   TABLOCK
);

BULK INSERT salaries
FROM 'C:\SQLData\salaries.csv'
WITH
```

```
(
    FIRSTROW = 2,
    FIELDTERMINATOR = ',',
    ROWTERMINATOR = '0x0a',
    TABLOCK
);
```

Problem Statement

Performance Analytics

1.Top 10 High Performing Employees

```
SELECT TOP 10
    e.emp_id,
    e.emp_name,
    d.dept_name,
    AVG(pr.rating) AS avg_rating
FROM performance_reviews pr
JOIN employees e ON pr.emp_id = e.emp_id
JOIN departments d ON e.dept_id = d.dept_id
GROUP BY e.emp_id, e.emp_name, d.dept_name
ORDER BY avg_rating DESC;
```

	emp_id	emp_name	dept_name	avg_rating
1	36	Priya Sharma 36	Finance	4.700000
2	56	Ananya Rao 56	Finance	4.700000
3	76	Priya Sharma 76	Finance	4.700000
4	96	Karthik R 96	HR	4.700000
5	26	Sneha Iyer 26	IT	4.700000
6	66	Rohit Das 66	IT	4.700000
7	6	Priya Sharma 6	Marketing	4.700000
8	86	Rohit Das 86	Marketing	4.700000
9	16	Divya Patel 16	Operations	4.700000
10	46	Priya Sharma 46	Operations	4.700000

2.Average Rating Per Department

```
SELECT
    d.dept_name,
    AVG(pr.rating) AS avg_department_rating
FROM performance_reviews pr
JOIN employees e ON pr.emp_id = e.emp_id
JOIN departments d ON e.dept_id = d.dept_id
GROUP BY d.dept_name
ORDER BY avg_department_rating DESC;
```

	dept_name	avg_department_rating
1	HR	3.844444
2	Operations	3.664705
3	Finance	3.661904
4	Marketing	3.538095
5	IT	3.491304

3.Employees Eligible for Promotion

```
SELECT
    e.emp_id,
    e.emp_name,
    d.dept_name,
    AVG(pr.rating) AS avg_rating,
    DATEDIFF(YEAR, e.join_date, GETDATE()) AS experience_years
FROM employees e
JOIN performance_reviews pr ON e.emp_id = pr.emp_id
JOIN departments d ON e.dept_id = d.dept_id
GROUP BY e.emp_id, e.emp_name, d.dept_name, e.join_date
HAVING
    AVG(pr.rating) > 4
    AND DATEDIFF(YEAR, e.join_date, GETDATE()) >= 2;
```

	emp_id	emp_name	dept_name	avg_rating	experience_years
1	4	Rohit Das 4	Finance	4.500000	8
2	34	Rohit Das 34	Finance	4.500000	8
3	36	Priya Sharma 36	Finance	4.700000	6
4	56	Ananya Rao 56	Finance	4.700000	3
5	76	Priya Sharma 76	Finance	4.700000	6
6	79	Rohit Das 79	Finance	4.200000	9

4.Performance Trend Over Years

```
SELECT
    review_year,
    AVG(rating) AS avg_rating
FROM performance_reviews
GROUP BY review_year
ORDER BY review_year;
```

	review_year	avg_rating
1	2020	3.630000
2	2021	3.630000
3	2022	3.630000
4	2023	3.630000
5	2024	3.630000
6	2025	3.630000
7	2026	3.630000
8	2027	3.630000
9	2028	3.630000
10	2029	3.630000

Salary Analytics

5.Salary distribution per department

```
select
    d.dept_name,
    count(s.salary) as total_employees,
    min(s.salary) as min_salary,
    max(s.salary) as max_salary,
    avg(s.salary) as avg_salary
from salaries s
join employees e on s.emp_id = e.emp_id
join departments d on e.dept_id = d.dept_id
group by d.dept_name
order by d.dept_name;
```

	dept_name	total_employees	min_salary	max_salary	avg_salary
1	Finance	21	51500.00	99500.00	77023.809523
2	HR	18	53000.00	97500.00	72805.555555
3	IT	23	50000.00	98500.00	76869.565217
4	Marketing	21	52500.00	99000.00	76547.619047
5	Operations	17	50500.00	98000.00	68911.764705

6.Highest and lowest salary per role

```
select
   jr.role_name,
   min(s.salary) as lowest_salary,
   max(s.salary) as highest_salary
from salaries s
join employees e on s.emp_id = e.emp_id
join job_roles jr on e.role_id = jr.role_id
group by jr.role_name
order by jr.role_name;
```

	role_name	lowest_salary	highest_salary
1	Analyst	50000.00	96000.00
2	Developer	52000.00	96500.00
3	Executive	50500.00	94500.00
4	Manager	52500.00	99000.00
5	Team Lead	51000.00	99500.00

7.Salary vs performance correlation

```
select
   e.emp_id,
   e.emp_name,
   avg(pr.rating) as avg_rating,
   s.salary
from employees e
join performance_reviews pr on e.emp_id = pr.emp_id
join salaries s on e.emp_id = s.emp_id
group by e.emp_id, e.emp_name, s.salary
order by avg_rating desc;
```

Results | Messages

	emp_id	emp_name	avg_rating	salary
1	6	Priya Sharma 6	4.700000	52500.00
2	16	Divya Patel 16	4.700000	57500.00
3	26	Sneha Iyer 26	4.700000	62500.00
4	36	Priya Sharma 36	4.700000	67500.00
5	46	Priya Sharma 46	4.700000	72500.00
6	56	Ananya Rao 56	4.700000	77500.00
7	66	Rohit Das 66	4.700000	82500.00
8	76	Priya Sharma 76	4.700000	87500.00
9	86	Rohit Das 86	4.700000	92500.00
10	96	Karthik R 96	4.700000	97500.00
11	94	Divya Patel 94	4.500000	96500.00
12	84	Sneha Iyer 84	4.500000	91500.00

8.Employees below department average salary

```
select
   e.emp_id,
   e.emp_name,
   d.dept_name,
   s.salary
from employees e
join salaries s on e.emp_id = s.emp_id
join departments d on e.dept_id = d.dept_id
where s.salary < (
   select avg(s2.salary)
   from employees e2
   join salaries s2 on e2.emp_id = s2.emp_id
   where e2.dept_id = e.dept_id
);
```

	emp_id	emp_name	dept_name	salary
1	7	Divya Patel 7	HR	53000.00
2	11	Sneha Iyer 11	HR	55000.00
3	17	Rohit Das 17	HR	58000.00
4	19	Rohit Das 19	HR	59000.00
5	23	Meera Nair 23	HR	61000.00
6	24	Sneha Iyer 24	HR	61500.00
7	27	Meera Nair 27	HR	63000.00
8	29	Sneha Iyer 29	HR	64000.00
9	40	Ananya Rao 40	HR	69500.00
10	41	Arun Kumar 41	Finance	70000.00
11	30	Vikram Singh 30	Finance	64500.00
12	34	Rohit Das 34	Finance	66500.00

Department Growth

9.Hiring trend per year

```
select
   year(join_date) as hire_year,
   count(*) as total_hired
from employees
group by year(join_date)
order by hire_year;
```

	hire_year	total_hired
1	2015	15
2	2016	9
3	2017	11
4	2018	10
5	2019	9
6	2020	10
7	2021	7
8	2022	11
9	2023	8
10	2024	10

10.Headcount growth by department

```
select
   d.dept_name,
   count(e.emp_id) as total_employees
from employees e
join departments d on e.dept_id = d.dept_id
group by d.dept_name
order by d.dept_name;
```

	dept_name	total_employees
1	Finance	21
2	HR	18
3	IT	23
4	Marketing	21
5	Operations	17

11.Attrition rate by department

```
select d.dept_name,sum(case when e.status = 'active' then 1 else 0 end) as active_employees,
sum(case when e.status = 'inactive' then 1 else 0 end) as inactive_employees
from employees e
join departments d on e.dept_id = d.dept_id
group by d.dept_name
order by d.dept_name;
```

	dept_name	active_employees	inactive_employees
1	Finance	7	14
2	HR	9	9
3	IT	13	10
4	Marketing	10	11
5	Operations	14	3

Strategic Insights

12.Employee overdue for promotion

```
select
   e.emp_id,
   e.emp_name,
```

```
    d.dept_name,
    avg(pr.rating) as avg_rating,
    datediff(year, e.join_date, getdate()) as experience_years
from employees e
join performance_reviews pr on e.emp_id = pr.emp_id
join departments d on e.dept_id = d.dept_id
group by e.emp_id, e.emp_name, d.dept_name, e.join_date
having datediff(year, e.join_date, getdate()) >= 3
and avg(pr.rating) >= 4
order by avg_rating desc;
```

	emp_id	emp_name	dept_name	avg_rating	experience_years
1	36	Priya Sharma 36	Finance	4.700000	6
2	56	Ananya Rao 56	Finance	4.700000	3
3	76	Priya Sharma 76	Finance	4.700000	6
4	26	Sneha Iyer 26	IT	4.700000	4
5	66	Rohit Das 66	IT	4.700000	4
6	86	Rohit Das 86	Marketing	4.700000	7
7	46	Priya Sharma 46	Operations	4.700000	8
8	16	Divya Patel 16	Operations	4.700000	9
9	14	Priya Sharma 14	Operations	4.500000	4
10	54	Priya Sharma 54	Operations	4.500000	5

13.High salary but low performance cases

```
select
    e.emp_id,
    e.emp_name,
    s.salary,
    avg(pr.rating) as avg_rating
from employees e
join salaries s on e.emp_id = s.emp_id
join performance_reviews pr on e.emp_id = pr.emp_id
group by e.emp_id, e.emp_name, s.salary
having s.salary > (select avg(salary) from salaries)
   and avg(pr.rating) < 3;
```

	emp_id	emp_name	salary	avg_rating
1	53	Rahul Verma 53	76000.00	2.800000
2	58	Arun Kumar 58	78500.00	2.500000
3	63	Arun Kumar 63	81000.00	2.800000
4	68	Ananya Rao 68	83500.00	2.500000
5	73	Vikram Singh 73	86000.00	2.800000
6	78	Rohit Das 78	88500.00	2.500000
7	83	Karthik R 83	91000.00	2.800000
8	88	Rahul Verma 88	93500.00	2.500000
9	93	Arun Kumar 93	96000.00	2.800000
10	98	Ananya Rao 98	98500.00	2.500000

14.Most stable department (lowest attrition)

```
select top 1
   d.dept_name,
   count(case when e.status = 'inactive' then 1 end) * 100.0
      / count(*) as attrition_rate
from employees e
join departments d on e.dept_id = d.dept_id
group by d.dept_name
order by attrition_rate asc;
```

	dept_name	attrition_rate
1	Operations	17.647058823529

15.Location-wise salary comparison

```
select
   l.location_name,
   avg(s.salary) as avg_salary,
   min(s.salary) as min_salary,
   max(s.salary) as max_salary
from employees e
join salaries s on e.emp_id = s.emp_id
join locations l on e.location_id = l.location_id
group by l.location_name
order by avg_salary desc;
```

	location_name	avg_salary	min_salary	max_salary
1	Chennai	78120.689655	52500.00	99000.00
2	Hyderabad	76902.777777	51500.00	99500.00
3	Bangalore	69742.857142	50000.00	97000.00

16.Gender diversity ratio

```
select
   gender,
   count(*) * 100.0 / (select count(*) from employees) as percentage
from employees
group by gender;
```

	gender	percentage
1	Female	48.000000000000
2	Male	52.000000000000

17.Median salary calculation

```
select distinct
   percentile_cont(0.5) within group (order by salary)
   over () as median_salary
from salaries;
```

	median_salary
1	74750

18.Identify leadership pipeline candidates

```
select
   e.emp_id,
   e.emp_name,
   jr.role_name,
   avg(pr.rating) as avg_rating,
   datediff(year, e.join_date, getdate()) as experience_years
from employees e
join performance_reviews pr on e.emp_id = pr.emp_id
```

```
join job_roles jr on e.role_id = jr.role_id
group by e.emp_id, e.emp_name, jr.role_name, e.join_date
having avg(pr.rating) >= 4.5
  and datediff(year, e.join_date, getdate()) >= 5;
```

	emp_id	emp_name	role_name	avg_rating	experience_years
1	44	Vikram Singh 44	Analyst	4.500000	7
2	76	Priya Sharma 76	Analyst	4.700000	6
3	86	Rohit Das 86	Analyst	4.700000	7
4	46	Priya Sharma 46	Developer	4.700000	8
5	64	Priya Sharma 64	Developer	4.500000	5
6	94	Divya Patel 94	Developer	4.500000	10
7	4	Rohit Das 4	Executive	4.500000	8
8	16	Divya Patel 16	Executive	4.700000	9
9	34	Rohit Das 34	Executive	4.500000	8
10	36	Priya Sharma 36	Executive	4.700000	6